
When Adapter Training Hurts: Manifold Preservation for Vector Retrieval

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Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through lightweight adapters, global objectives produce dense gradient fields that systematically distort the latent manifold of unseen categories, causing out-of-distribution (OOD) retrieval performance to collapse below the frozen baseline. To address this, we propose **Euclidean Geodesic Alignment (EGA)**, a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient regime in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space—including regions occupied by unseen categories—structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings. On strictly unseen fine-grained categories, competing adapters degrade below the frozen CLIP baseline in label precision, whereas EGA improves over the strongest competing adapter by up to 12.5 percentage points.

1 Introduction

Modern retrieval systems built on foundation models such as CLIP [21] face a fundamental challenge: the pretrained encoder is general-purpose, but each downstream retrieval task requires embeddings tuned to its own category structure. Full fine-tuning is rarely viable due to the size of foundation backbones and the risk of destroying their zero-shot capabilities. The dominant solution is parameter-efficient adaptation through lightweight adapters trained on a labeled subset of the target task while the backbone remains frozen.

In practice, adapter training is almost universally driven by global contrastive objectives—methods that compute losses over the full pairwise similarity matrix in each mini-batch and continuously reshape the entire embedding distribution. Such objectives have proven effective for in-distribution classification, where all relevant categories appear during training. However, retrieval systems must also perform well on *unseen* categories that arrive at deployment time but were absent from adapter training. We show that in this out-of-distribution (OOD) regime, global contrastive adapters fail in a striking way: the dense gradient fields they generate during training do not stop at the seen-class boundary. Instead, they propagate throughout the embedding space, distorting the local manifold structure that the pretrained backbone had established for unseen categories. The empirical consequence is that adapters trained with state-of-the-art global objectives such as ICon [1] and SRL [4] retrieve unseen-class queries *less* accurately than the frozen baseline, i.e., adapter training actively harms OOD performance.

We propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that resolves this tension by replacing dense global optimization with sparse local updates. EGA combines three design choices: (i) a zero-initialized residual structure, (ii) an ℓ_2 normalization constraint, and (iii) a small-margin triplet loss on the class-aware neighborhood graph, each of which restricts the adapter’s influence to local neighborhood relationships that explicitly violate the retrieval objective. As training proceeds, the fraction of triplets producing non-zero gradient decays rapidly: at convergence, 96.5% of triplets contribute zero gradient. The adapter ceases to modify the vast majority of the embedding space, leaving the pretrained manifold structure for unseen categories intact.

We evaluate EGA across in-distribution (CIFAR-100) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101). On in-distribution data, EGA improves retrieval quality over the frozen baseline. On strictly unseen categories, EGA preserves and improves the pretrained manifold while ICon and SRL collapse below it: on FGVC-Aircraft, EGA reaches Label Precision 0.548 versus 0.423/0.441 for ICon/SRL, with the frozen baseline at 0.512. We trace this contrast to the gradient sparsity property and argue that it constitutes a general design principle for retrieval adapters built on foundation models.

1.1 Contributions

This paper makes the following contributions:

- **Phenomenon.** We identify a fundamental in-distribution/OOD tradeoff in global contrastive adapter training: while global objectives achieve higher geometric compression on seen categories, they produce dense gradient fields that systematically distort the pretrained manifold of unseen categories, causing retrieval performance to fall below the frozen CLIP baseline on out-of-distribution data.
- **Method.** We propose EGA, a lightweight residual adapter combining zero initialization, ℓ_2 normalization, and a small-margin triplet loss. By restricting optimization to locally violated neighborhood constraints, EGA refines the pretrained embedding space without destroying its global semantic structure, simultaneously improving retrieval quality on seen categories and preserving manifold integrity for unseen ones.
- **Mechanism and evaluation.** We provide a mechanistic explanation grounded in gradient sparsity: EGA’s triplet objective converges to a regime where 96.5% of training triplets contribute zero gradient, leaving unseen class regions structurally unchanged from CLIP. We validate this mechanism empirically and demonstrate through evaluations on three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101) that EGA outperforms ICon and SRL on both label precision and ANNS recall, while competing methods degrade below the frozen baseline.

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [21] produce embedding spaces whose structure encodes both seen-class semantics and a broader prior over visual concepts not observed during downstream training. A central concern in adapter tuning is whether this prior is preserved or destroyed during adaptation. Catastrophic forgetting has been studied extensively in continual and incremental learning settings, where successive task updates erode previously acquired representations [18, 31]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [36], structural fracturing under modality gap [22, 5], and loss of compositional structure under aggressive optimization [20, 24, 30]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [23] or restrict optimization to limited subspaces. However, these constraints typically operate at the global distribution level and apply uniform pressure across the embedding space. They do not distinguish between regions occupied by seen classes (where adaptation is needed) and regions occupied by unseen classes (where the pretrained structure should remain untouched). EGA addresses this distinction directly: rather than imposing global geometric constraints, it leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen-class regions of the pretrained manifold structurally unmodified.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [27, 17]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution, a strategy applied across various domains including graph encoders [28] and multimodal recommendation systems [37]. While pushing embeddings toward a uniform distribution enhances robustness to background noise [29], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like I Con formulate a unifying framework to systematically align multi view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [4]. Other approaches attempt to decouple the learning objectives [16] or apply information theoretic regularizers [35] to control the density of the learned representations. Furthermore, region based distillation techniques have been proposed to enhance fine grained understanding [13]. However, these global contrastive objectives and structural regularizers typically generate dense gradient fields throughout the training process. By continuously applying optimization pressure across the entire batch, they inadvertently disrupt the delicate pre established geometric structures of unseen fine grained categories. Unlike these global approaches that aggressively reconfigure the entire latent manifold through dense updates, EGA utilizes local geometric anchors to perform sparse and targeted metric smoothing, thereby preserving the structural integrity of out of distribution semantics.

Parameter Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [8, 9]. This paradigm has extensively shifted towards vision language models, dominating recent literature due to the prohibitive cost of full scale fine tuning [19]. Methods such as Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [34]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine tuning while preserving the generalizable representations of the foundation model [25]. To further protect the pre trained manifold during initial training steps, zero initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [33]. While these existing methods successfully maintain global knowledge and prevent catastrophic forgetting, they are not designed to resolve local metric inconsistencies for vector search. Unlike these general adaptation strategies, EGA integrates a zero initialized residual adapter with a localized geometric objective, explicitly smoothing class specific clusters to correct Euclidean distances without compromising the parameter efficiency or stability of the pre trained manifold.

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for retrieving high dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [14], while graph based indices and hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [26, 2]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [32] and analyzing the theoretical constructions of navigable graphs for high dimensional nearest neighbor search [3]. Furthermore, system level studies optimize retrieval latency and stability by aligning best first search algorithms with solid state drives [6] and enabling direct insertions in large scale indices [7]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low rank matrix factorization have also been introduced to accelerate inner product evaluations [12]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst case performance limitations [11]. While these sophisticated indexing and approximation systems continuously push the boundaries of retrieval throughput, they operate under the strict assumption that the underlying Euclidean space perfectly reflects semantic similarity. If the foundational manifold suffers from geometric distortion, these highly optimized engines will simply retrieve semantically irrelevant neighbors at an accelerated rate. Unlike hardware and system level optimizations that treat the embedding space as an immutable input, EGA proactively recalibrates the metric manifold prior to index construction, thereby enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

Label Precision@K measures the semantic quality of retrieved results—whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k -th nearest neighbor of q in the transformed embedding space.

ANNS Recall@K measures the geometric fidelity of the ANN index—the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget n_{probe} :

$$\text{AR@K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top- K neighbor sets, respectively.

The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@K by making the geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4, global contrastive objectives achieve high AR@K on seen classes but collapse on *both* metrics for unseen classes, whereas EGA improves both simultaneously under the OOD setting.

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is deliberately constrained by three design principles, each of which serves a distinct role in preserving the pretrained manifold.

Residual connection with zero initialization. Each EGA block applies a residual update of the form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation (Table 2), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

Feature gating. Inside each residual block, the transformed signal passes through a feature-gating module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

where $W_1 \in \mathbb{R}^{(d/4) \times d}$, $W_2 \in \mathbb{R}^{d \times (d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise multiplication. This structure, analogous to Squeeze-and-Excitation networks [10], learns a per-dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones.

187 The bottleneck ratio of $1/4$ is chosen to prevent the gating network from overfitting on seen-class
188 data, as validated in Table 3.

189 **ℓ_2 normalization.** The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2
190 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

191 This constraint ensures that the transformed embeddings remain in the same metric space as the
192 original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable
193 on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage
194 points (Table 2), confirming that hypersphere consistency is necessary for stable ANN indexing.

195 **Full forward pass.** Let Block_i denote the i -th EGA block consisting of a two-layer MLP with Lay-
196 erNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm.
197 The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) . \quad (6)$$

198 The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension
199 2048, which is negligible compared to the frozen CLIP backbone ($\sim 150M$ parameters).

200 3.3 Local Triplet Training and Gradient Sparsity

201 **Training objective.** Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a
202 is an anchor image, p is a positive sampled uniformly from the same class as a , and n is a negative
203 sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

204 where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the
205 transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis
206 in Table 3.

207 **Gradient sparsity and manifold preservation.** A triplet (a, p, n) contributes a non-zero gradient
208 to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

209 We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many
210 triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 7 (left)
211 shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady
212 state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast
213 majority of local neighborhood relationships in the embedding space.

214 This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes
215 are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided
216 that the adapter’s updates remain local—i.e., that gradients from seen-class triplets do not propagate
217 into unseen-class regions of the embedding space—the CLIP manifold structure for unseen classes
218 is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class
219 neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of
220 actively-violated local relationships continue to be updated.

221 **Contrast with global contrastive objectives.** Global objectives such as ICon [1] and SRL [4]
222 compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a
223 KL divergence between the empirical similarity distribution and the class-membership distribution,
224 while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives
225 generate dense gradient fields in every training step: every sample pair—regardless of whether
226 its local geometry already satisfies the retrieval objective—contributes to the loss and receives a
227 non-zero gradient. This global optimization pressure from seen-class training propagates throughout
228 the embedding space, distorting the local manifold structure that CLIP had established for unseen
229 categories.

Proposition 1 (Gradient Sparsity at Convergence). *Let f_θ be trained with the triplet loss in Eq. (7) with margin $m > 0$ on a dataset $\mathcal{D}_{\text{train}}$ drawn from seen classes $\mathcal{Y}_{\text{seen}}$. Define the active triplet ratio at training step t as*

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} \left[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \right]. \quad (9)$$

As f_θ converges, $\rho(t) \rightarrow \rho^*$, where

$$\rho^* = \Pr_{(a,p,n)} \left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \quad (10)$$

is determined solely by the margin m and the within-class distance distribution of the converged embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.

Proof Sketch. The loss in Eq. (7) is a sum of hinge terms, each of which has gradient zero whenever $d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m \leq 0$. At convergence, f_θ has minimized the loss over $\mathcal{D}_{\text{train}}$, so the only remaining active triplets are those whose geometry the adapter has not yet been able to separate within the margin. The fraction of such triplets is bounded below by the overlap of the within-class and between-class distance distributions at the converged embedding, which depends on m and the geometry of $\mathcal{Y}_{\text{seen}}$ but not on $\mathcal{Y}_{\text{unseen}}$. Since no triplet involving unseen-class embeddings is ever constructed, the adapter gradient is identically zero over the entire unseen-class region of the embedding space throughout training. See Appendix A for the complete proof. $\square$

The empirical validation of Proposition 1 is provided in Section 4.4, where we track the active triplet ratio throughout training and connect it directly to the OOD retrieval outcomes shown in Figure 7.

4 Evaluation

4.1 Experimental Setup

Experiment Platform All experiments were conducted on the Chameleon Cloud [15] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.

4.2 In-Distribution Validation

We first verify that EGA functions as a competent retrieval adapter under the standard in-distribution setting before turning to the out-of-distribution claims that constitute the paper’s main contribution. All experiments in this section use CIFAR-100, with the adapter trained and evaluated on the same 100-class label space.

Semantic retrieval quality. Figure 1 reports Label Precision@ K across four neighborhood sizes and three search budgets. EGA improves Label Precision@1 from ≈ 0.55 (raw CLIP) to > 0.70 at $n_{\text{probe}} = 1$, and the gap persists across all K and n_{probe} values. The improvement is not an artifact of larger search budgets: even at $n_{\text{probe}} = 10$, EGA retains a clear advantage, indicating that the adapter genuinely tightens within-class neighborhoods rather than merely benefiting from broader index coverage.

Geometric structure. The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top- K neighbors against random background points, before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K , with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall–efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.67 Recall@1

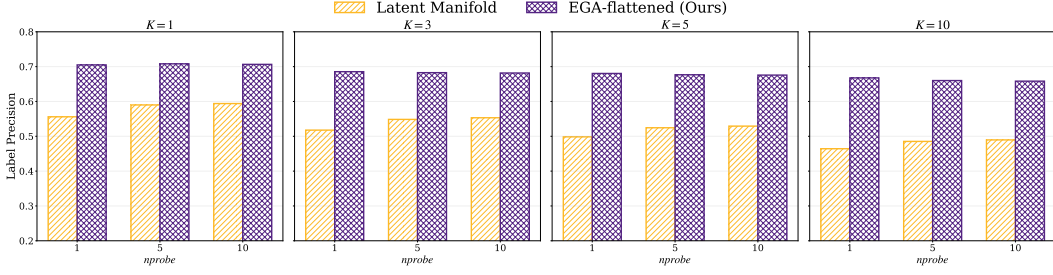


Figure 1: Label Precision@ K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and $nprobe \in \{1, 5, 10\}$. EGA consistently improves over the raw CLIP latent space.

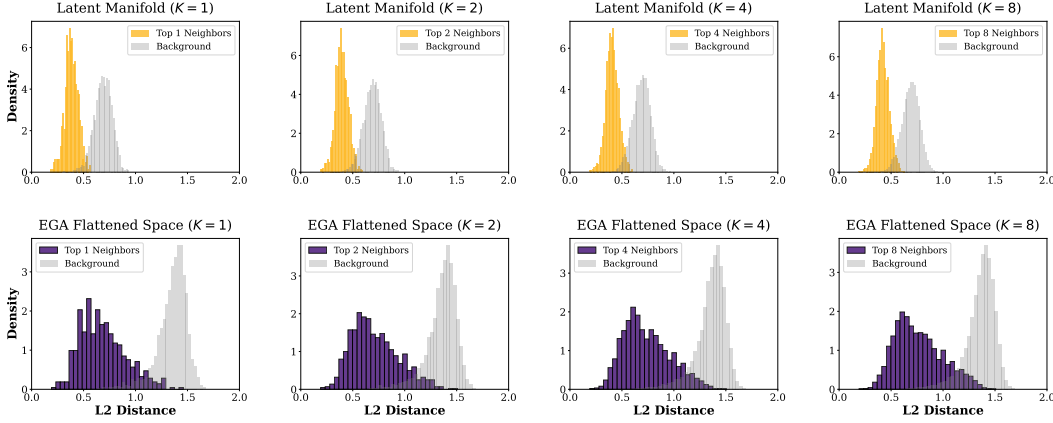


Figure 2: L_2 distance distributions for Top- K neighbors versus random background points on CIFAR-100. The raw CLIP space exhibits significant overlap; EGA separates the two distributions cleanly.

at $nprobe = 1$, whereas EGA reaches 0.80 under identical indexing, and saturates above 0.97 by $nprobe = 5$.

These in-distribution results establish a baseline competence: EGA improves both semantic and geometric retrieval quality on the categories it was trained on. The more important question—how it behaves on categories absent from training—is the subject of the remaining subsections.

4.3 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper: under strict OOD settings where evaluation categories are disjoint from training categories, EGA simultaneously improves both retrieval metrics, while global contrastive adapters harm at least one of them relative to the frozen baseline. Table 1 summarizes the headline numbers across all three OOD benchmarks at $K = 1$, $nprobe = 1$.

The generalization capabilities of EGA are evaluated on the CIFAR-10 dataset, providing a rigorous out-of-distribution (OOD) test since all models were trained exclusively on CIFAR-100. Figure 4 illustrates the recall performance across varying $nprobe$ values and K settings. EGA consistently maintains a dominant lead over all baselines, including recent state-of-the-art methods like ICon [1] and SRL [4].

At a minimum search budget of $nprobe=1$, EGA achieves a Recall@1 of 0.858, significantly outperforming SRL (0.831) and ICon (0.788). As shown in the inset zooms, EGA continues to hold a clear advantage even at $nprobe=10$, where other models begin to saturate. A key observation is the efficiency gain provided by our method: EGA achieves a recall level at $nprobe=5$ that matches or exceeds the performance of supervised baselines at $nprobe=10$. This indicates that the manifold smoothing provided by EGA effectively doubles the search efficiency in unseen data environments by creating a more index-friendly geometric structure.

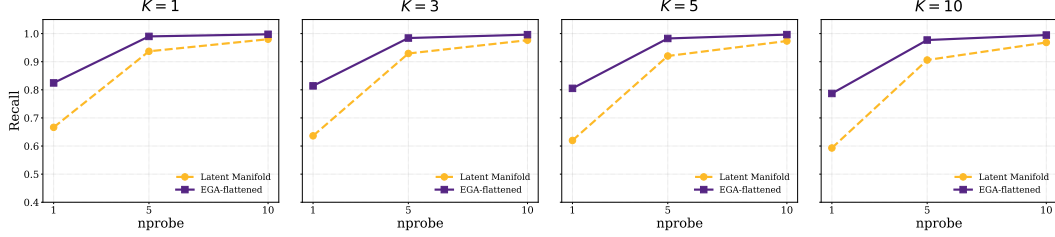


Figure 3: ANNS Recall@ K versus nprobe on CIFAR-100. EGA shifts the efficiency frontier upward across all K .

Table 1: Out-of-distribution retrieval performance at $K = 1$, nprobe = 1. EGA is the only adapter that improves both Label Precision and ANNS Recall over the frozen CLIP baseline across all three OOD settings. Bold marks values strictly above the frozen CLIP baseline.

Dataset	Metric	CLIP	ICon	SRL	EGA (Ours)
CIFAR-10	ANNS Recall@1	0.617	0.788	0.831	0.858
	Label Precision@1	—	—	—	—
FGVC-Aircraft	ANNS Recall@1	0.773	0.875	0.863	0.898
	Label Precision@1	0.512	0.423	0.441	0.548
Food-101	ANNS Recall@1	—	—	—	—
	Label Precision@1	0.842	0.838	0.840	0.897

294 To further validate the robustness of EGA under fine-grained domain shifts, we extend our evaluation
295 to the FGVC Aircraft dataset. Figure 5 visually corroborates that EGA remains the only method
296 capable of scaling zero-shot retrieval performance on fine-grained domains without destroying the
297 underlying manifold. In this setting, models are trained on 80 known aircraft categories and tested on
298 20 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training
299 identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter
300 advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial
301 preservation capabilities. Global contrastive and structural regularization methods, such
302 as ICon and SRL, suffer severe performance degradation when confined to a lightweight adapter
303 bottleneck. Specifically, at $K = 1$ and nprobe=1, the recall for ICon and SRL collapses to 0.422
304 and 0.440 respectively, falling significantly below the original frozen CLIP baseline of 0.773. This
305 phenomenon indicates that global contrastive objectives aggressively distort the latent manifold of
306 unseen fine-grained categories when the representation capacity is constrained. Conversely, EGA
307 exhibits exceptional stability and generalization, elevating the recall to 0.898 under the exact same
308 adapter constraints. By leveraging local geometric anchors rather than forcing global distributions,
309 EGA successfully compresses class-specific clusters while maintaining the structural integrity of the
310 broader semantic space.

311 To comprehensively evaluate the versatility of EGA across diverse fine-grained domains, we extend
312 our evaluation to the Food-101 dataset. Figure 6 visually corroborates that EGA remains the only
313 method capable of scaling zero-shot retrieval performance on fine-grained domains without destroying
314 the underlying manifold. In this setting, models are trained on 80 known food categories and tested
315 on 21 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training
316 identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter
317 advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial
318 preservation capabilities, particularly given the strong pre-trained representations of common food
319 items. Global contrastive and structural regularization methods, such as ICon and SRL, suffer
320 performance degradation when confined to a lightweight adapter bottleneck. Specifically, at $K = 1$
321 and nprobe=1, the recall for ICon and SRL drops to 0.838 and 0.840 respectively, falling below
322 the original frozen CLIP baseline of 0.842. This phenomenon indicates that global contrastive
323 objectives aggressively distort the already well-structured latent manifold of unseen categories when
324 the representation capacity is constrained. Conversely, EGA exhibits exceptional stability and
325 generalization, elevating the recall to 0.897 under the exact same adapter constraints. By leveraging
326 local geometric anchors rather than forcing global distributions, EGA successfully compresses class-
327 specific clusters while maintaining the structural integrity of the broader semantic space, significantly
328 enhancing retrieval efficiency even on a highly optimized baseline manifold.

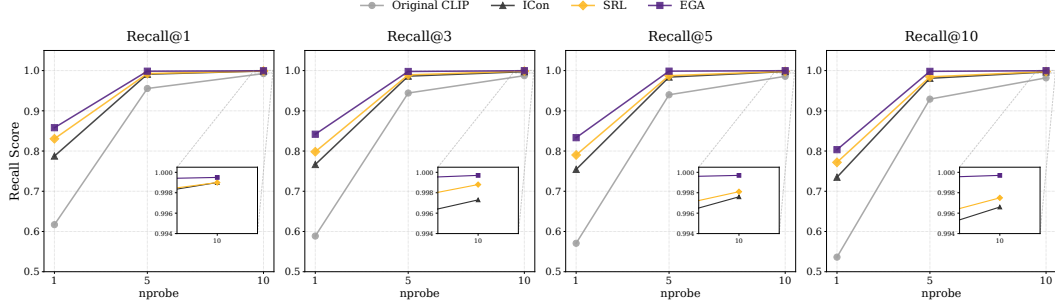


Figure 4: Recall@K versus nprobe tradeoff on CIFAR-10. EGA maintains a consistent lead over baselines, demonstrating strong OOD generalization.

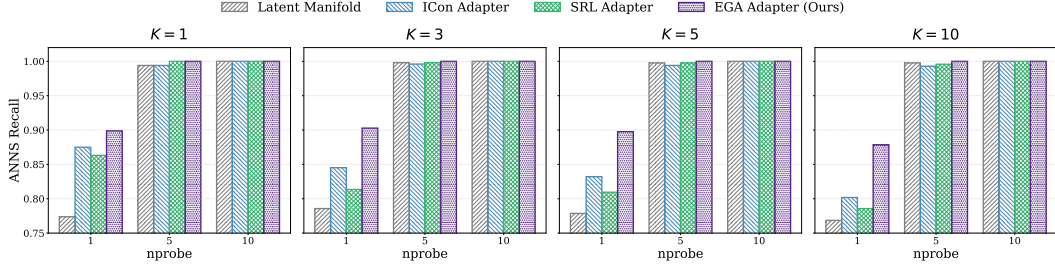


Figure 5: Recall versus nprobe tradeoff on 20 unseen FGVC Aircraft classes.

329 4.4 Mechanistic Analysis: Gradient Sparsity as OOD Protection

330 The superior OOD generalization of EGA raises a fundamental question: why does a triplet-trained
 331 adapter preserve unseen class structure while global contrastive objectives destroy it? We identify
 332 **gradient sparsity** as the key mechanism.

333 A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the
 334 margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

335 As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 7
 336 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by conver-
 337 gence. At steady state, **96.5% of triplets contribute zero gradient**, meaning the adapter updates
 338 only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding
 339 space—including regions occupied by unseen classes—structurally intact.

340 Global contrastive objectives such as ICon [1] and SRL [4] exhibit the opposite behavior. ICon
 341 minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes
 342 batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step:
 343 every sample pair contributes to the loss regardless of whether its local geometry already satisfies
 344 the retrieval objective. This global optimization pressure shapes the entire embedding distribution
 345 based on seen-class similarities, even for regions of the space occupied by unseen-class samples. The
 346 result is not loss of geometric structure—in fact, ICon and SRL produce highly compact clusters on
 347 unseen data—but rather loss of *semantically correct* structure: unseen samples are pulled toward the
 348 clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward
 349 other unseen samples that share their true category.

350 Figure 7 (right) quantifies the consequence on FGVC Aircraft, where all models are trained on 80 seen
 351 classes and evaluated strictly on 20 unseen classes. ICon and SRL collapse to label precision of 0.423
 352 and 0.441, respectively—**below the frozen CLIP baseline of 0.512**—confirming that dense gradient
 353 updates actively harm zero-shot retrieval. EGA, by contrast, improves to 0.548 while modifying only
 354 3.5% of the embedding geometry at convergence.

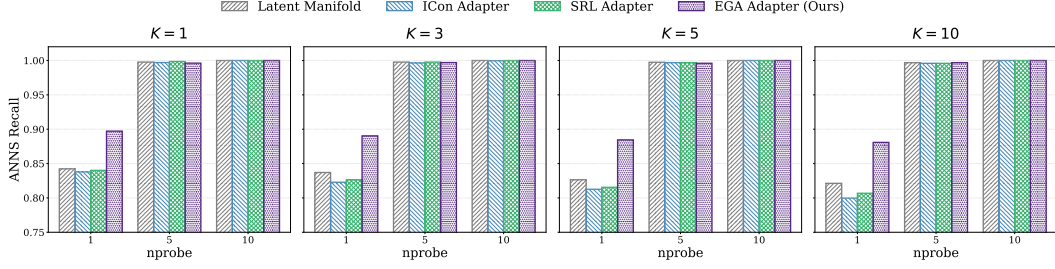


Figure 6: Recall versus nprobe tradeoff on 21 unseen Food-101 classes.

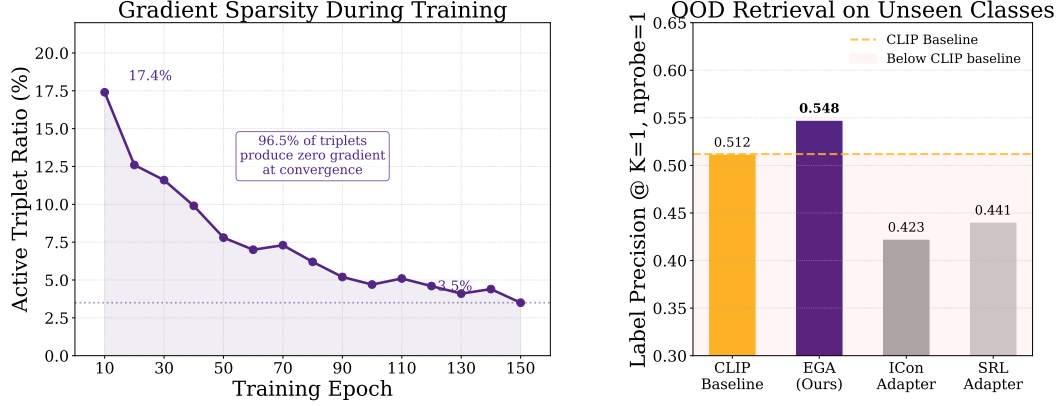


Figure 7: **Gradient sparsity as the mechanism behind EGA’s OOD generalization.** **Left:** Active triplet ratio during training on 80 seen FGVC Aircraft classes. By epoch 150, only 3.5% of triplets produce non-zero gradients, leaving 96.5% of the embedding space—including unseen class regions—structurally unchanged from CLIP. **Right:** Label Precision@1 at nprobe=1 on 20 strictly unseen classes. ICon and SRL, which produce dense gradient fields throughout training, both collapse *below* the frozen CLIP baseline entirely, suggesting that adapter training under global objectives can actively harm zero-shot retrieval rather than simply failing to improve it. EGA’s sparse updates preserve the pretrained manifold and improve retrieval on unseen categories.

355 This analysis points to a key distinction in adapter design for foundation models: **dense global**
 356 **objectives reshape the embedding space according to seen-class similarities, projecting unseen**
 357 **samples onto the resulting structure; sparse local objectives leave the pretrained semantic**
 358 **relationships among unseen samples intact.** The contrast is not merely one of degree—ICon
 359 and SRL fall below the frozen CLIP baseline entirely, suggesting that adapter training under global
 360 objectives can actively harm zero-shot retrieval rather than simply failing to improve it.

361 4.5 Ablation Study

362 We perform a comprehensive ablation study on the CIFAR 100 dataset to verify the effectiveness of
 363 our design components. The results are summarized in Table 2. Each architectural choice contributes
 364 to the structural integrity of the calibrated manifold.

365 The residual connection is the most essential part of the design. When the residual connection is
 366 removed, the accuracy collapses from 0.7015 to 0.0065. This failure proves that the EGA adapter
 367 should refine the existing semantic space rather than replace it. Without this shortcut, the lightweight
 368 MLP cannot reconstruct the complex semantic relationships from limited training data.

369 L2 normalization is also necessary for stable search performance. Its removal leads to a performance
 370 loss of approximately 4.2 percent. This confirms that the hypersphere constraint is vital for consistent
 371 distance metrics in vector similarity search. Finally, the zero initialization strategy provides a
 372 1.7 percent gain. By starting the adapter as an identity mapping, the model preserves pre trained
 373 knowledge at the beginning of the process, which provides a superior starting point for manifold
 374 smoothing.

Table 2: Ablation study of EGA components on CIFAR-100.

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

4.6 Hyperparameter Sensitivity Analysis

The internal stability of EGA is evaluated by varying the triplet margin m and the hidden dimension d of the adapter. As presented in Table 3, our method demonstrates a consistent performance plateau across a broad range of settings.

The results for $K = 5$ indicate that EGA maintains high retrieval quality even when the margin m fluctuates. While the $K = 1$ metric shows sensitivity to larger margin values, the overall stability of the manifold is preserved. This trend suggests that a moderate margin setting, specifically between 0.1 and 0.3, is optimal for calibrating fine grained latent spaces. Furthermore, the performance remains nearly identical as the hidden dimension d scales from 512 to 4096. This observation confirms that the geometric correction provided by EGA is structural in nature and does not rely on massive parameter counts.

Table 3: Sensitivity of retrieval performance to different hyperparameter configurations on CIFAR 100.

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

5 Conclusion

We have shown that the standard practice of training retrieval adapters with global contrastive objectives carries a previously unexamined cost: it actively harms retrieval performance on categories absent from adapter training. Through controlled comparisons on FGVC Aircraft, we demonstrated that state-of-the-art methods such as ICon and SRL push label precision on unseen classes *below* the frozen CLIP baseline, while reshaping the embedding space according to seen-class similarities.

Our proposed adapter, EGA, addresses this failure mode through a different optimization regime rather than a different architecture. By combining a zero-initialized residual structure, ℓ_2 normalization, and a small-margin triplet loss, EGA converges to a sparse-gradient regime in which 96.5% of training triplets contribute zero gradient at steady state. Across CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101, EGA improves both Label Precision and ANNS Recall over both the frozen baseline and the strongest competing adapters, with the largest gains appearing exactly where global methods fail—on strictly unseen categories.

The broader implication of our analysis is that not all retrieval improvements are robust to distribution shift. Methods that compress seen-class clusters most aggressively are not necessarily the methods that generalize best. We suggest that gradient sparsity at convergence is a useful diagnostic for predicting how an adapter will behave on unseen data, and that adapter design for foundation models should be evaluated under explicit OOD settings rather than in-distribution benchmarks alone.

Limitations. Our analysis has three notable limitations. First, our empirical study focuses exclusively on visual representations from CLIP; while the gradient sparsity mechanism is loss-function-specific rather than modality-specific, validation on text and multimodal retrieval (e.g., Sentence-BERT, BLIP) remains future work. Second, EGA requires labeled supervision on seen categories to construct the class-aware triplet graph; fully unsupervised variants that derive positives from

the pretrained kNN structure are an open direction. Third, our gradient sparsity analysis (Proposition 1) characterizes what happens to unseen regions assuming they remain disjoint from seen-class neighborhoods in the embedding space; when seen and unseen categories overlap heavily in the pretrained metric, even sparse local updates can affect unseen samples indirectly. Quantifying this overlap-dependence is a question for future analysis.

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867 A Proof of Proposition 1

868 *Proof.* We prove each claim in turn.

869 **(1) Gradient support.** The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$, where
 870 $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

871 which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_{\theta} \mathcal{L}$ receives contributions only from
 872 active triplets satisfying Eq. (8).

873 **(2) Convergence of $\rho(t)$.** Under mild regularity conditions (Lipschitz continuity of f_{θ} and bounded
 874 embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity,
 875 $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding
 876 distribution and the embeddings converge.

877 **(3) Zero gradient over unseen regions.** Let $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$ denote the region of the embedding
 878 space occupied by unseen-class images under f_{θ} . By construction, all triplets (a, p, n) in $\mathcal{D}_{\text{train}}$ satisfy
 879 $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$, so $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$ for all training triplets. The gradient $\nabla_{\theta} \delta_{apn}$ depends on
 880 f_{θ} only through its evaluation at $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$, which lie outside $\mathcal{R}_{\text{unseen}}$. Consequently, no training step
 881 produces a gradient that directly modifies f_{θ} in $\mathcal{R}_{\text{unseen}}$, and the CLIP geometry in that region is
 882 preserved throughout training.

883 **(4) Dependence on m .** The residual distance gap δ_{apn}^* at convergence is bounded by the within-class
 884 diameter $\Delta_c = \max_{x, x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c . For $m \leq \min_c \Delta_c$, the adapter can drive all
 885 within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets
 886 remain active at convergence because the within-class spread exceeds the margin, resulting in a
 887 strictly positive ρ^* . This explains the empirically observed increase in ρ^* with larger m reported in
 888 Table 3. $\square$